



## Week 7 - Graded Assignment 7



**The due date for submitting this assignment has passed.**

Due on 2026-04-01, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

**You have last submitted on: 2026-04-01, 07:55 IST**

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

1 point

Consider the vector spaces  $V$  and functions  $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$  defined as follows:

i)  $V = \mathbb{R}^2$  and  $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2$ .

ii)  $V = M_{2 \times 2}(\mathbb{R})$  and  $\langle A, B \rangle = \text{Tr}(AB)$ , where  $\text{Tr}(M)$  denotes the trace of a matrix  $M$ , i.e., the sum of the diagonal elements of  $M$ .

iii)  $V = M_{2 \times 1}(\mathbb{R})$  and  $\langle A, B \rangle = \text{Tr}(AB^t)$ , where  $\text{Tr}(X)$  denotes the trace of a matrix  $X$ , i.e., the sum of the diagonal elements of  $X$ .  $Y^t$  denotes the transpose of matrix  $Y$ .

iv)  $V = \mathbb{R}^2$  and  $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 + x_2 y_1$ .

- ☒ (i) is an inner product.
- ☒ (ii) is an inner product.
- ☐ (iii) is an inner product.
- ☐ (iv) is an inner product.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

- (i) is an inner product.
- (iii) is an inner product.

1 point

Consider two linear transformations  $T$  and  $S$  from  $\mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined as  $T(x, y) = (2x + y, x + y)$  and  $S(x, y) = (x + cy, x + 2y)$ . Let  $A$  and  $B$  be matrix representations of linear transformations  $T$  and  $S$  with respect to the standard bases of  $\mathbb{R}^2$  respectively. Consider the following statements:

- P:** If  $c = 1$ , then  $A$  and  $B$  are similar matrices.
- Q:** If  $c = 2$ , then  $A$  and  $B$  are similar matrices.
- R:** If  $c = 1$  and  $P^{-1}AP = B$ , then  $P$  can be the matrix  $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ .
- S:** If  $c = 1$  and  $P^{-1}AP = B$ , then  $P$  can be the matrix  $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ .
- T:** If  $c = 1$ , then there are infinitely many  $P$  satisfying the equation  $P^{-1}AP = B$ .

Which of the following options are true?

- ☒  $P$  is true but  $Q$  is false.
- ☐



Both  $P$  and  $QP$  and  $Q$  are true.



Both  $R$  and  $SR$  and  $S$  are true.



$RR$  is false but  $SS$  is true.



$TT$  is true.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$PP$  is true but  $QQ$  is false.

$RR$  is false but  $SS$  is true.

$TT$  is true.

1 point

Let  $\langle \cdot, \cdot \rangle$  denote the standard inner product on  $\mathbb{R}^2$ , i.e.,  $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ . Which one of the following options is (are) true for the vector  $\gamma \in \mathbb{R}^2$ , such that  $\langle \alpha, \gamma \rangle = 4$  and  $\langle \beta, \gamma \rangle = 8$ , where  $\alpha = (3, 1)$  and  $\beta = (6, 2)$ .



No such  $\gamma$  exists.



There are infinitely many such vectors which satisfy the properties of  $\gamma$ .



$\gamma$  is unique in  $\mathbb{R}^2$ .



Any vector in the set  $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$  satisfies the properties of  $\gamma$ .



$(1, 1)$  is the only vector which satisfies the properties of  $\gamma$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

There are infinitely many such vectors which satisfy the properties of  $\gamma$ .

Any vector in the set  $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$  satisfies the properties of  $\gamma$ .

1 point

Let  $\langle \cdot, \cdot \rangle$  denote the standard inner product on  $\mathbb{R}^2$ , i.e.,  $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$  and let  $v \in \mathbb{R}^2$ . Consider a linear transformation  $T_v : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined as:

$$T_v(u) = \langle u, v \rangle v,$$

where  $v \in \mathbb{R}^2$ . Which of the following options is (are) true for  $T_v$ ?



$T_v$  is one-one for all  $v \neq 0 \in \mathbb{R}^2$ .



$T_v$  is onto for all  $v \neq 0 \in \mathbb{R}^2$ .



$T_v$  is onto for all  $v \in \mathbb{R}^2$ .



$T_v$  is not one-one for every  $v \in \mathbb{R}^2$ .



There exists a  $v \in R^2$   $v \in R^2$  such that  $T_v$   $T_v$  is not onto.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$T_v$   $T_v$  is onto for all  $v \neq 0 \in R^2$   $v \neq 0 \in R^2$ .

$T_v$   $T_v$  is not one-one for every  $v \in R^2$   $v \in R^2$ .

There exists a  $v \in R^2$   $v \in R^2$  such that  $T_v$   $T_v$  is not onto.

1 point

Let  $A$  and  $B$  be square matrices of the same order  $n$ . Which of the following statements are sufficient to conclude that  $A$  is equivalent to  $B$ .

☐  $\det(A) = \det(B)$ .

☒  $\text{Nullity}(A) = \text{Nullity}(B)$ .

☐

The set of columns of  $AA$ , and the set of columns of  $BB$  are both linearly dependent subsets of  $R^n$   $R^n$ .

☒

There exists a matrix  $CC$  such that  $AA$  is equivalent to  $CC$ , and  $BB$  is equivalent to  $CC$ .

☒

There exist invertible matrices  $PP$  and  $QQ$  of order  $n$  such that  $PA = BQ$   $PA = BQ$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$\text{Nullity}(A) = \text{Nullity}(B)$ .

There exists a matrix  $CC$  such that  $AA$  is equivalent to  $CC$ , and  $BB$  is equivalent to  $CC$ .

There exist invertible matrices  $PP$  and  $QQ$  of order  $n$  such that  $PA = BQ$   $PA = BQ$ .

1 point

Let  $L$  and  $L'$  be affine subspaces of  $R^3$   $R^3$ , where  $L = (0, 1, 1) + U$   $L = (0, 1, 1) + U$  and  $L' = (0, 1, 0) + U'$   $L' = (0, 1, 0) + U'$ , for some vector subspaces  $U$  and  $U'$  of  $R^3$   $R^3$ . Let a basis for  $U$  be given by  $\{(1, 1, 0), (1, 0, 1)\}$   $\{(1, 1, 0), (1, 0, 1)\}$  and a basis for  $U'$  be given by  $\{(1, 0, 0)\}$   $\{(1, 0, 0)\}$ . Suppose there is a linear transformation  $T: U \rightarrow U'$   $T: U \rightarrow U'$  such that  $(1, 0, 1) \in \ker(T)$   $(1, 0, 1) \in \ker(T)$  and  $T(1, 1, 0) = (1, 0, 0)$   $T(1, 1, 0) = (1, 0, 0)$ . An affine mapping  $f: L \rightarrow L'$   $f: L \rightarrow L'$  is obtained by defining  $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$   $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$ , for all  $u \in U$   $u \in U$ . Which of the following options are true?

☒

$L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$   $L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$ .

☐

$L' = \{(x, y + 1, 0) \mid x, y \in R\}$   $L' = \{(x, y + 1, 0) \mid x, y \in R\}$ .

☐

$L = \{(x - y, y + 1, x - y + 1) \mid x, y \in R\}$   $L = \{(x - y, y + 1, x - y + 1) \mid x, y \in R\}$ .

☐

$L = \{(x, x + 1, y + 1) \mid x, y \in R\}$   $L = \{(x, x + 1, y + 1) \mid x, y \in R\}$ .

☒

$f(x, y + 1, x - y + 1) = (y, 1, 0)$   $f(x, y + 1, x - y + 1) = (y, 1, 0)$

☐

$f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$   $f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$

☐

$f(x, x + 1, y + 1) = (y, 1, 0)$   $f(x, x + 1, y + 1) = (y, 1, 0)$

☐

$f(x, y + 1, x - y + 1) = (0, 1, y)$   $f(x, y + 1, x - y + 1) = (0, 1, y)$

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

$$L = \{(x, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\} L = \{(x, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\}.$$

$$f(x, y + 1, x - y + 1) = (y, 1, 0) f(x, y + 1, x - y + 1) = (y, 1, 0)$$

Let  $\theta$  be the angle between the vectors  $u = (4, 7, 3) u = (4, 7, 3)$  and  $v = (1, 2, -6) v = (1, 2, -6)$ , then what will be the value of  $\cos(\theta) \cos(\theta)$ ?

0

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

(Type: Numeric) 0

1 point

Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\} \{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\}$$

of  $\mathbb{R}^3$  with usual inner product. Suppose  $v = (x, y, \frac{3x+y}{5}) \in V v = (x, y, 53x+y) \in V$  is written as  $v = c_1 v_1 + c_2 v_2 + c_3 v_3 v = c_1 v_1 + c_2 v_2 + c_3 v_3$ , such that  $c_1 + c_2 = 4 c_1 + c_2 = 4$ . What will be the value of  $c_3 c_3$ ?

2

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

(Type: Numeric) 2

1 point

Let  $V = \mathbb{R}^2 V = \mathbb{R}^2$  be a vector space. Consider two inner products  $\langle \cdot, \cdot \rangle_1$  and  $\langle \cdot, \cdot \rangle_2$  on  $V$  defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4 y_1 y_2 \langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4 y_1 y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3 x_1 x_2 + 2 y_1 y_2 \langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3 x_1 x_2 + 2 y_1 y_2.$$

If  $\langle (a, b), (4, 3) \rangle_1 = 35 \langle (a, b), (4, 3) \rangle_1 = 35$  and  $\langle (a, b), (4, 3) \rangle_2 = 60 \langle (a, b), (4, 3) \rangle_2 = 60$ , then find the value  $a + 2b a + 2b$ .

11

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

(Type: Numeric) 11

1 point

Let  $V = \mathbb{R}^3 V = \mathbb{R}^3$  be the inner product space with usual inner product. If  $\theta$  is the angle between  $(15, 8, 18) (15, 8, 18)$  and  $(a, b, c) (a, b, c)$  where  $15a + 8b + 18c = 0 15a + 8b + 18c = 0$  and  $(a^2 + b^2 + c^2) \neq 0 (a^2 + b^2 + c^2) \neq 0$ , then find the value of  $\cos \theta \cos \theta$ .

0

**Yes, the answer is correct.**

**Score: 1**

### Accepted Answers:

(Type: Numeric) 0

1 point

Consider a vector  $\mathbf{v} = (x - 20, 2, 1) \in \mathbb{R}^3$ . Find the value of  $x$  so that the length of the vector  $\mathbf{v}$  is minimum.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 20

1 point

Consider the following subsets of  $\mathbb{R}^3$ , with  $\langle \cdot, \cdot \rangle$  used to denote the standard dot product on  $\mathbb{R}^3$ , and  $\|\cdot\|$  used to denote the standard length function.

$S_1 = \{\mathbf{v} \in \mathbb{R}^3 \mid \langle \mathbf{v}, (0, 0, 1) \rangle = 2, \|\mathbf{v}\| = 1\}$   
 $S_2 = \{\mathbf{u} \in \mathbb{R}^3 \mid \mathbf{u} \in \text{Span}(\{(1, 1, 1)\}), \|\mathbf{u}\| = 1\}$   
 $S_1 = \{\mathbf{v} \in \mathbb{R}^3 \mid \langle \mathbf{v}, (0, 0, 1) \rangle = 2, \|\mathbf{v}\| = 1\}$   
 $S_2 = \{\mathbf{u} \in \mathbb{R}^3 \mid \mathbf{u} \in \text{Span}(\{(1, 1, 1)\}), \|\mathbf{u}\| = 1\}$

If  $n_1$  and  $n_2$  denote the number of elements in  $S_1$  and  $S_2$ , respectively, then find the value of  $n_1 - n_2$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) -2

1 point